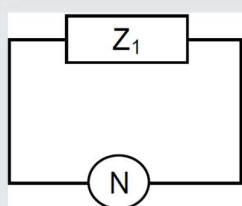
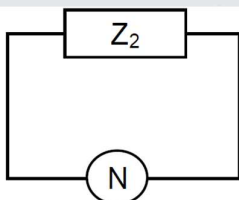


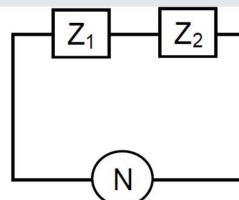
1. In the first AC circuit current from the AC source is $I_1 = I_0 \sin(\omega t - 37^\circ)$, in the second AC circuit current from AC source is $I_2 = I_0 \sin(\omega t + 53^\circ)$, current from AC source in third AC circuit will be :



$$E = E_0 \sin \omega t$$

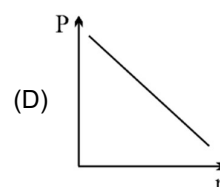
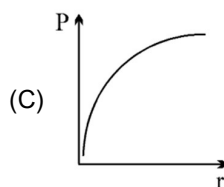
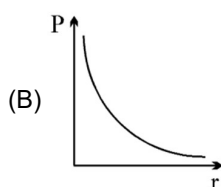
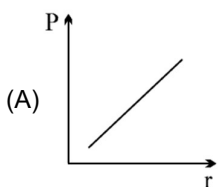


$$E = E_0 \sin \omega t$$

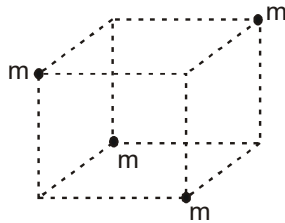


$$E = E_0 \sin \omega t$$

- (A) $\frac{I_0}{\sqrt{2}} \sin\left(\omega t + \tan^{-1}\left(\frac{1}{7}\right)\right)$
- (B) $\frac{I_0}{\sqrt{2}} \sin\left(\omega t + \tan^{-1}\left(\frac{2}{7}\right)\right)$
- (C) $\frac{I_0}{2} \sin\left(\omega t + \tan^{-1}\left(\frac{2}{7}\right)\right)$
- (D) $\frac{I_0}{2} \sin\left(\omega t + \tan^{-1}\left(\frac{1}{7}\right)\right)$
2. Two soap bubbles of radius 3 cm and 4 cm coalesce to form a single bubble under isothermal conditions. Then the radius of bigger bubble is (Taking surrounding pressure equal to zero)
- (A) 7 cm (B) 1 cm (C) 12 cm (D) 5 cm
3. Consider two uniformly charged concentric and coaxial rings of radii R and $2R$. Total charge on inner ring is Q_1 and that on outer ring is Q_2 . Both rings are revolving in same sense with same angular velocity about its axis. If net magnetic induction at a distance R from the centre of the rings, on axis of rings is zero then is :
- (A) -1 (B) $-\frac{2\sqrt{2}}{5\sqrt{5}}$ (C) $-\frac{8\sqrt{2}}{5\sqrt{5}}$ (D) $-\frac{4\sqrt{2}}{3\sqrt{3}}$
4. A spherical soap bubble placed in vacuum is blown such that its radius increases at a constant rate. Which of following curves represents power required to increase surface energy of the bubble versus radius of drop.

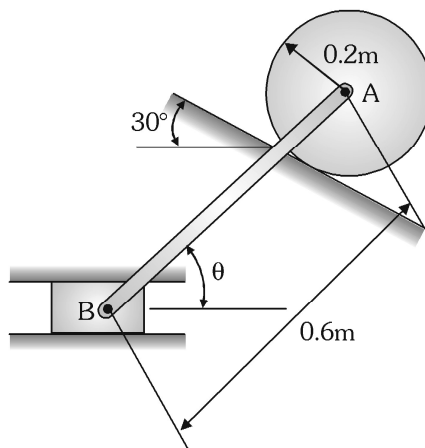


5. Four point masses each of mass m are kept at four corners of a cube of side length ' a '. Then



- (A) the gravitational potential energy of system is $= -\frac{3\sqrt{2}Gm^2}{a}$.
- (B) the gravitational potential energy of system is $= -\frac{4\sqrt{2}Gm^2}{a}$.
- (C) the magnitude of force on one of the particle due to the remaining masses is equal to $\frac{\sqrt{6}Gm^2}{a^2}$.
- (D) the magnitude of force on one of the particle due to others is equal to $\sqrt{\frac{3}{2}} \frac{Gm^2}{a^2}$.

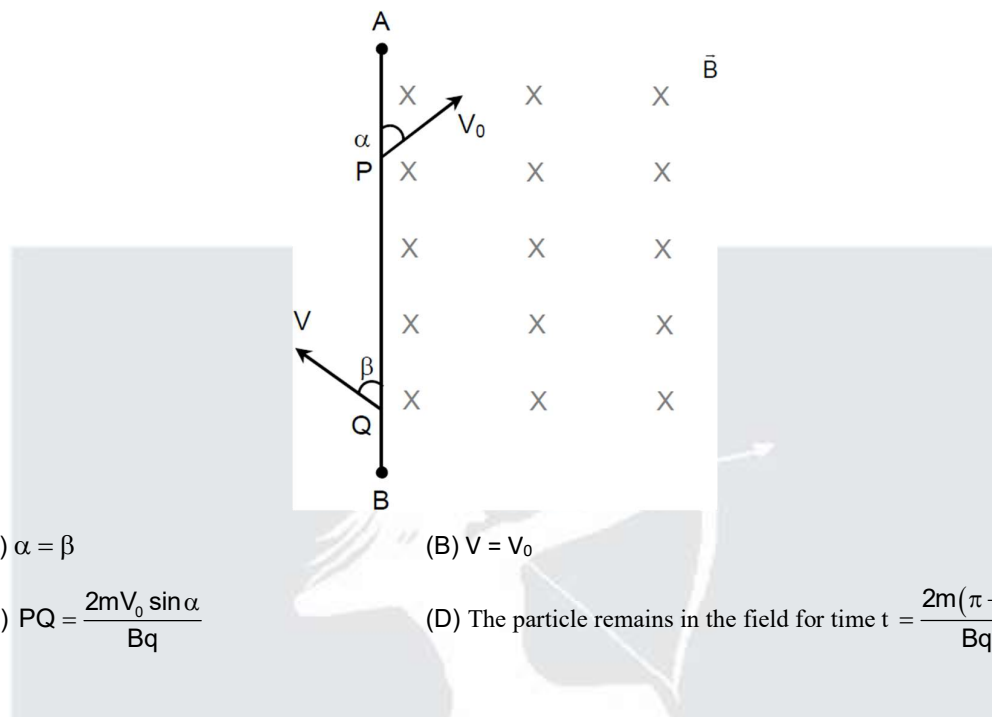
6. The disk is released from rest and rolls down the incline. Knowing that the speed of A is 1.2 m/s when $\theta = 0^\circ$, at that instant (Only portions of the two tracks are shown)



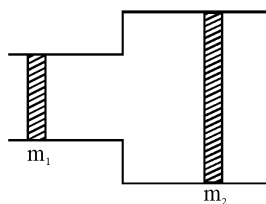
- (A) Angular velocity of the rod is 9 rad/s
- (B) Angular velocity of the rod is 1 rad/s
- (C) The velocity of B is 1.039 m/s
- (D) The velocity of B is 7.29 m/s

Power of real gurus

7. There exist a uniform magnetic field perpendicular to plane of the paper (inward) right side of the line AB. A particle of charge $-q$ and mass m enters a uniform magnetic field $\vec{B}$ (perpendicular to paper inward) at P with a velocity V_0 at an angle α and leaves the field at Q with velocity V at angle β as shown in figure.

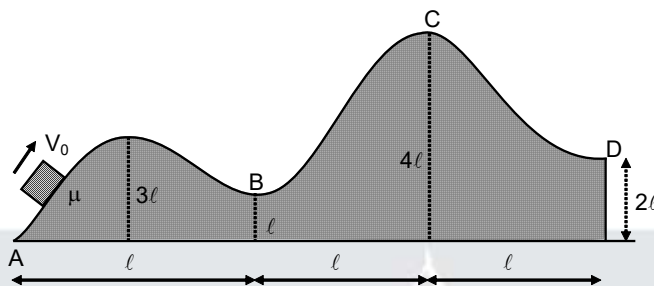


- (A) $\alpha = \beta$
 (B) $V = V_0$
 (C) $PQ = \frac{2mV_0 \sin \alpha}{Bq}$
 (D) The particle remains in the field for time $t = \frac{2m(\pi - \alpha)}{Bq}$
8. Two tubes with cross sections S_1 and S_2 are interconnected and closed by the pistons, whose masses are equal to m_1 and m_2 respectively. An explosion occurs in the gas enclosed. Some time after explosion first piston was moving with speed v to the left and v_2 is the speed of other piston. Friction of the piston tube wall can be neglected then :-

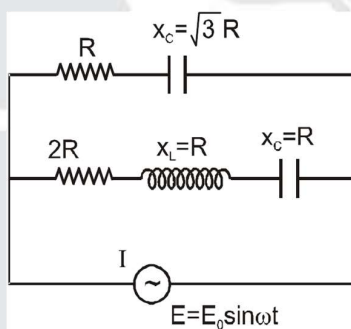


- (A) If the pipe is fixed and cannot move; $v_2 = m_1 v / m_2$
 (B) If the pipe is not fixed and its mass is M ; $v_2 = m_1 v / (m_2 + m_1 + M)$
 (C) If the pipe is fixed and cannot move; $v_2 = m_1 S_2 v / m_2 S_1$
 (D) If the pipe is not fixed and its mass is M then $v_{\text{pipe}} = \frac{m_1 v (S_2 - S_1)}{S_1 M}$

9. A block is moved very slowly from A to D with the help of a variable force which is always tangential to the surface of the fixed curve path. Coefficient of friction between block and surface is μ . (The block always remain in contact with plane). Then choose the correct alternative(s) :



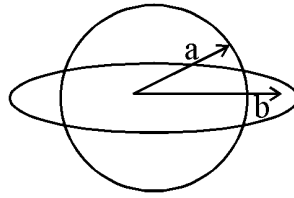
- (A) Work done by friction from A to D is $-3\mu mg\ell$
- (B) If $\mu = 1$, work done by variable force on block from A to C is $6mg\ell$
- (C) If $\mu = 1$, work done by gravity force on block from B to C is $-3mg\ell$
- (D) If $\mu = 1$, work done by variable force on block from A to D is $5mg\ell$
10. Choose the correct options regarding the AC circuit given in the figure: (Symbols have their usual meaning)



- (A) Equivalent impedance of the circuit is $\frac{2R}{\sqrt{3}}$
- (B) Power factor of the circuit is $\frac{\sqrt{3}}{2}$
- (C) Average power for complete cycle is $\frac{3E_0^2}{8R}$
- (D) Effective reactance of the circuit is $\frac{R}{\sqrt{3}}$
11. A small plane can fly at a speed of 150 km/h in still air. A 50 km/h wind is blowing from west to east. Plane flies a distance 500 km due north. Choose the correct options :

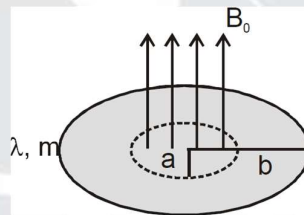
- (A) Time taken by the plane is $\frac{5}{\sqrt{2}}h$
- (B) Time taken by the plane is $\frac{10}{3}h$
- (C) Speed of plane is $100\sqrt{2}$ km/h
- (D) Plane should fly at an angle $\theta = \sin^{-1}\left(\frac{1}{3}\right)$ west of north relative to wind.

12. An insulating ring of charge $2\pi\lambda b$, radius b is concentric with a charged conducting solid sphere of charge Q and radius a ($b > a$) as shown in the figure. Then



- (A) Potential of the sphere is $\frac{Q}{4\pi\epsilon_0 a} + \frac{\lambda}{2\epsilon_0}$
- (B) If sphere is grounded then charge on the sphere will be $\frac{-a\lambda}{8\pi^2\epsilon_0}$
- (C) Electric field inside the sphere is non-zero
- (D) The sphere is equipotential body.

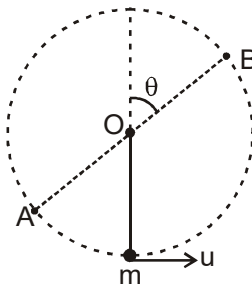
13. A nonconducting ring of uniform mass m , radius b and uniform linear charge density ' λ ' is suspended as shown in figure in a gravity free space. There is uniform coaxial magnetic field B_0 , pointing up in a circular region of radius ' a ' ($< b$). Now if this field is switched off, then :-



- (A) There will be induced electric field on periphery of ring, in anticlockwise sense when seen from above
- (B) Induced electric field imparts angular momentum of magnitude $\lambda\pi a^2 b B_0$
- (C) Final angular velocity of ring will be more if time taken to switch off the field (B_0) is small
- (D) Final angular velocity will always be independent of time taken to switch off the field (B_0).

PARAGRAPH

A ball is hanging vertically by a light inextensible string of length L from fixed point O . The ball of mass m is given a speed u at the lowest position such that it completes a vertical circle with centre at O as shown. Let AB be a diameter of circular path of ball making an angle θ with vertical as shown. (g is acceleration due to gravity)



14. Let T_A and T_B be tension in string when ball is at A and B respectively, then $T_A - T_B$ is equal to

(A) $6 mg \cos\theta$ (B) $6 mg$ (C) $12 mg \cos\theta$

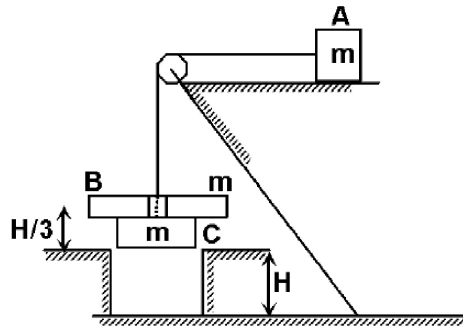
(D) None of these

15. Let $\vec{a}_A$ and $\vec{a}_B$ be acceleration of ball when it is at A and B respectively, then $|\vec{a}_A + \vec{a}_B|$ is equal to

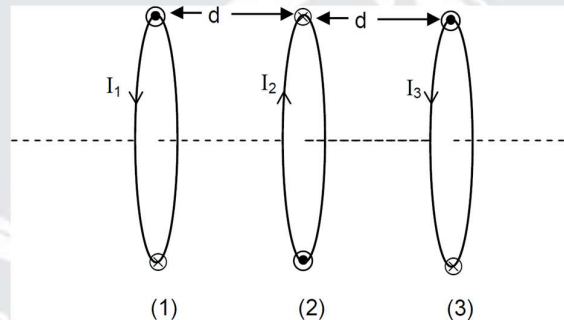
(A) $2g \sin\theta$ (B) $g\sqrt{12\cos^2\theta + 4}$ (C) $4g \cos\theta$

(D) None of these

16. The system in figure is released from rest from the position shown. After blocks have moves distance $H/3$, collar B is removed and block A and C continue to move. The speed of C just before it strikes the ground is $\sqrt{\frac{13gH}{n}}$. Find n.



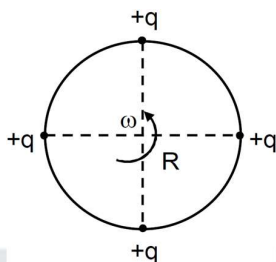
17. Consider 3 identical conducting loops kept parallel to each other such that their axes coincide. Loop (1), loop (2) and loop (3) self inductance of each loop is L . Current in loop (1) is I_1 , in loop (2) I_2 and in loop (3) I_3 , rate of change of current in them at certain instant is $\frac{dI_1}{dt}$, $\frac{dI_2}{dt}$, $\frac{dI_3}{dt}$ respectively. Their mutual inductance is M_{ij} (i.e. for 1 & 2 its M_{12}) current in loop (1) and (3) is in same sense and in (2) its in opposite sense :



Given that if $\frac{dI_1}{dt} = 2 \frac{dI_2}{dt} = \frac{dI_3}{dt}$, emf induced in loop (2) is zero. Find $\frac{L}{M_{12}}$

18. A particle A is moving in xy plane with the constant speed of $2\pi m/s$ along the path $x^2 + y^2 + 120y = 0$. At time $t = 0$. When A is at the origin, another particle B starts moving from origin with constant acceleration is such a way that at time $t = 5s$ velocities of both the particles are found to be equal. If sense of motion of A is clockwise, the average speed of B over first five second is $n\pi m/s$. Find n.
19. Four identical charges are fixed on the periphery of a non conducting ring of radius R as shown. Ring is rotated with constant angular velocity ω w.r.t. an axis perpendicular to plane of the ring and passing

through the centre of the ring. Force acting on any of the charge particle only due to the magnetic interaction between the charges is $F = \frac{\mu_0 q^2 \omega^2}{4\alpha\pi} \left[\sqrt{\alpha} + \frac{1}{\sqrt{\alpha}} \right]$ Here α is an integer. Find the value of α .



20. A resistance (R), inductance (L) and capacitance (C) are connected in series to an ac source of voltage v having variable frequency. If the energy delivered by the source to the circuit during one period if the operating frequency is twice the resonance frequency is $\frac{\pi R \sqrt{LC} V^2}{R^2 + P \left(\frac{L}{C} \right)}$. Find R

ANSWER KEY

- | | | | | |
|------------|-----------|----------|-----------|---------|
| 1. (D) | 2. (D) | 3. (C) | 4. (A) | 5. (AD) |
| 6. (BC) | 7. (ABCD) | 8. (CD) | 9. (ABCD) | |
| 10. (ABCD) | 11. (ACD) | 12. (AD) | 13. (ABD) | 14. (A) |
| 15. (B) | 16. 9 | 17. 4 | 18. 1 | 19. 2 |
| 20. 2.25 | | | | |